

RESTful Booker API — Software Testing Report

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1. Project Overview

This report documents functional, negative, and basic load testing performed on the RESTful Booker REST API. The objective was to verify authentication, booking creation and retrieval, update operations, negative scenarios, and API behaviour under a small concurrent load.

2. Test Environment & Tools

Application	RESTful Booker API
API Base URL	https://restful-booker.herokuapp.com
API Testing Tool	Postman
Load Testing Tool	Apache JMeter
Load Test	10 threads × 5 loops = 50 samples

3. Functional / API Test Results

ID	Endpoint	Scenario	Result
TC-01	POST /auth	Valid authentication	PASS
TC-02	POST /booking	Create new booking	PASS
TC-03	GET /booking/{id}	Retrieve booking	PASS
TC-04	GET /booking/999999	Invalid booking ID returns 404	PASS
TC-05	PUT /booking/999999	Invalid booking update returns 404	PASS
TC-06	PUT /booking/3386	Update existing booking	PASS
TC-07	GET /booking/3386	Verify updated booking persists	PASS
TC-08	GET /booking	Retrieve all bookings	PASS
TC-09	POST /auth	Invalid credentials	OBSERVATION
TC-10	DELETE /booking/3386	Delete booking	BLOCKED / OBSERVATION
TC-11	GET /booking?firstname=...	Booking filtering	OBSERVATION

4. Key Functional Test Details

- **Authentication:** Valid credentials returned an authentication token successfully.
- **Booking creation:** A booking was created successfully and returned a booking ID.
- **Booking retrieval:** The created booking data was successfully retrieved.
- **Booking update:** Booking ID 3386 was updated successfully with HTTP 200, and a subsequent GET verified the updated values.
- **Negative testing:** Requests using a non-existent booking ID returned HTTP 404 as expected.
- **Invalid authentication:** Incorrect credentials returned HTTP 200 with a 'Bad credentials' response. This was recorded as an observation rather than a confirmed defect because the API specification was not independently

verified.

- **Delete operation:** DELETE returned HTTP 403 during testing. It was recorded as blocked/observation rather than a confirmed defect because the authentication/specification behaviour was not independently verified.
- **Filtering:** Filtering attempts did not return the expected booking data consistently. This was recorded as an observation because the public shared test environment can change data and the endpoint specification was not independently verified.

5. Load Testing with JMeter

A small-scale load test was executed against GET /booking using Apache JMeter. The test used 10 concurrent threads, a 5-second ramp-up period, and 5 loops per thread, producing 50 total samples.

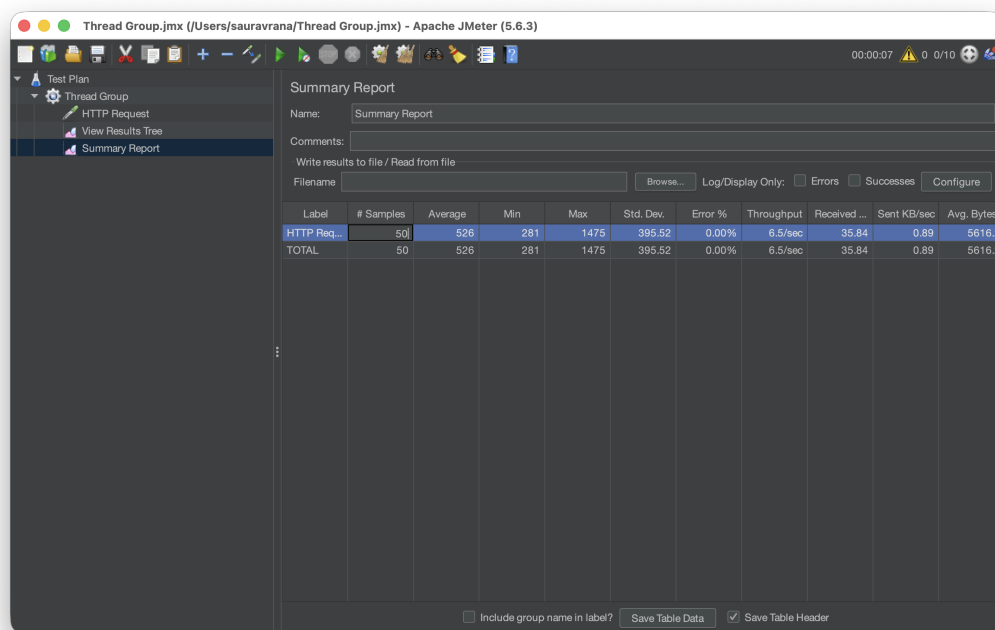


Figure 1: JMeter Summary Report — 50 samples, 0.00% error rate

Metric	Observed Result
Samples	50
Average Response Time	526 ms
Minimum Response Time	281 ms
Maximum Response Time	1475 ms
Error Percentage	0.00%
Throughput	6.5 requests/sec
Standard Deviation	35.84 ms

The load test completed with 0% reported errors for the 50 samples. Because the test was small-scale and executed against a public shared API, these measurements should not be treated as a full capacity or stress benchmark.

6. Summary

Area	Result
Confirmed functional PASS tests	8
Observations / blocked scenarios	3
Load test samples	50
Load test error rate	0.00%

7. Conclusion

The RESTful Booker API was tested using Postman for functional and negative scenarios and Apache JMeter for a basic load test. The main booking operations tested successfully, while several behaviours were documented as observations or blocked rather than classified as confirmed defects due to the limitations of the public shared test environment and lack of independent specification verification.

Note: JMeter Summary Report screenshot included above in Section 5.